

David Zhou

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EDUCATION

McGill University

Expected December 2027

Bachelor of Software Engineering Co-op

CGPA: 3.71 / 4.0

- Machine Learning, Algorithms & Data Structures, Algorithm Design, Computer Architecture, Operating Systems, Database Systems, Software Engineering Practice, Software Validation, Model-Based Programming

EXPERIENCE

Amazon Web Services

May 2026 – August 2026

Software Development Engineer Intern

- Owned a custom endpoint feature for IAM Identity Center, replacing system URLs with customer-owned domains
- Shipped 6 Coral APIs integrating CloudFront and Route53, scaling to **200,000 tenants** per shared distribution
- Implemented latency-based failover with automated health checks, reducing regional outage recovery time to **~90s**
- Engineered an async Step Function poller to track CloudFront's **~5 min** edge propagation status with timeout

StreamingFast

May 2025 – August 2025

Software Engineer Intern

- Contributed to protocol infrastructure using Go, improving data streaming speed in distributed systems
- Optimized firehose tracing of transactions using concurrency, reducing real-time streaming latency by **40%**
- Led migration of critical data pipelines from P2P architecture to gRPC, increasing syncing speed by **350%**
- Reduced I/O latency by executing operations in RAM disk, increasing data throughput from **5MB/s to 200MB/s**

National Research Council

May 2024 – August 2024

Research Assistant Intern

- Developed computer vision systems for medical devices with a **0.05 mm** error margin, improving surgical safety
- Automated performance testing infrastructure with Python and OpenCV, enabling **4x faster** deployment cycles
- Implemented 3D object tracking functionality on Raspberry Pi, reducing hardware cost by **\$200 per unit**
- Collaborated with Queen's University researchers and medical teams to ensure alignment of client needs

PROJECTS

Polymarket Trading Bot

- Designed an autonomous prediction bot that executes trades on Polymarket, generating **15%** in realized profit
- Deployed a multi-threaded dispatch system with per-algorithm worker isolation and **5%** slippage gate
- Integrated a high-frequency polling engine screening **500,000+** real daily trades in the market

TECHNICAL SKILLS

Languages: Python, Go, Java, C, Bash, JavaScript, HTML, CSS

Frameworks: Spring Boot, Junit, OpenCV, gRPC, AWS SDK, Flask, Protocol Buffers

Tools: Git, Github, Docker, Vim, AWS, Postman, REST API, JWT, Jira, SQL, NoSQL

Concepts: Backend, Frontend, Fullstack, Infrastructure, Software Engineering, API, Automation, CI/CD, Agile, OOP

ACHIEVEMENTS

Varsity Athlete

- Bronze medal – Jeux du Québec, the **Under-18 Olympic Games of Quebec**
- Ranked **top 200** in Quebec with a 1h19 finish in the half marathon